

SQL DATA ANALYST PORTFOLIO PROJECT

ZEPTO E-COMMERCE SQL DATA ANALYST PORTFOLIO PROJECT

READY FOR A BUSINESS LENS

Start with structure before interpretation.

The dataset is ready for a business lens

CORE TABLE

A SKU-level table named *zepto*, with category, product name, MRP, discount, quantity, weight, stock status, and inventory quantity.

FIRST CHECKS

Count rows, inspect sample records, find null product names, list distinct categories, compare in-stock and out-of-stock products, and surface duplicate product names.

```
SELECT COUNT(*) FROM zepto;
```

Price fields need cleaning before they can guide a decision

The cleaning step is a trust layer between raw data and commercial insight.

01

Remove invalid prices

Identify rows where MRP or discounted selling price equals zero, then delete rows with zero MRP.

MRP = 0

02

Normalize units

Divide MRP and discounted selling price by 100 to convert paise into rupees.

price / 100

03

Trust layer

Show the cleaning step before any value, revenue, or category interpretation.

raw → clean → decide

Eight questions turn raw SKUs into an analysis plan

CUSTOMER VALUE

Three questions

Top discounts.

High-MRP products that are out of stock.

Products above ₹500 with discounts below 10%.

COMMERCIAL PERFORMANCE

Two questions

Estimated revenue by category.

The five categories with the highest average discount.

OPERATIONAL VALUE

Three questions

Price per gram.

Weight bands.

Total inventory weight by category.

Discount and availability expose customer-facing trade-offs

Q1

Best-value products

Rank the ten best-value products by discount percentage.

ORDER BY discount % DESC

Q2

Availability gap

Isolate products with MRP above ₹300 that are currently out of stock.

MRP > ₹300 · out of stock

Q4

Low-discount, high-price

Find products above ₹500 with discounts under 10%, ordered by MRP and discount.

MRP > ₹500 · discount < 10%

PAIR THE
SIGNALS

Pair promotion depth with availability so “best value” does not ignore whether a shopper can actually buy the item.

Revenue and unit economics need category context

ESTIMATED REVENUE

Category scale

Sum discounted selling price multiplied by available quantity for each category.

$\text{selling price} \times \text{available quantity}$

AVERAGE DISCOUNT

Promotion intensity

Calculate the mean discount percentage by category and rank the top five categories.

$\text{AVG}(\text{discount \%})$

PRICE PER GRAM

Product-level value

Compare products weighing at least 100 grams by discounted selling price divided by weight.

$\text{selling price} \div \text{weight}$

Separate category scale, promotion intensity, and product-level value instead of collapsing them into one metric.

SIZE MAKES INVENTORY OPERATIONAL

Storage, handling, and replenishment conversations start with visible weight.

Weight bands

01

Low

Below 1,000
grams

02

Medium

Below 5,000
grams

03

Bulk

Everything else

INVENTORY FOOTPRINT

Sum weight in grams multiplied by available quantity for each category.

$\text{weight} \times \text{available quantity}$

PORTFOLIO STANDARD

QUERY TO DECISION

A credible analysis makes its path visible.

A strong portfolio project shows the path from query to decision

01

Start with trust

Inspect structure, nulls, duplicates, stock status, and invalid prices.

02

Make units comparable

Normalize paise to rupees before calculating value or revenue.

03

Translate SQL

Promotion, availability, category economics, unit value, and inventory footprint.

SCOPE NOTE

The source contains query logic, not returned result tables, so this presentation focuses on the analytical approach rather than fabricating findings.

Schema and data checks

```
-- Schema
drop table if exists zepto;

create table zepto(
sku_id SERIAL PRIMARY KEY,
category VARCHAR(120),
name VARCHAR(120),
mrp NUMERIC(8,2),
discountPercent NUMERIC(5,2),
availableQuantity INTEGER,
discountedSellingPrice NUMERIC(8,2),
weightInGms INTEGER,
outOfStock BOOLEAN,
quantity INTEGER
);
```

--Different Product Category

```
select DISTINCT category
from zepto
order by category;
```

--Count of rows

```
select COUNT(*) from zepto
```

--Sample data

```
select * from zepto LIMIT 10
```

--Null Values

```
select * from zepto
where name IS NULL
```

--Product in stock vs out of stock

```
select outOfStock, COUNT(sku_id)
from zepto
group by outOfStock
```

--Product names present multiple times

```
select name, COUNT(sku_id) as "Number of SKUs"
from zepto
group by name
```

Data cleaning

```
-- Data Cleaning
```

```
--product with price = 0
```

```
select * from zepto  
where mrp = 0 OR discountedSellingPrice = 0;
```

```
DELETE FROM zepto  
where mrp = 0;
```

```
--Convert paise to rupees
```

```
UPDATE zepto  
SET mrp = mrp/100.0,  
discountedSellingPrice =  
discountedSellingPrice/100.0;
```

```
select mrp, discountedSellingPrice from zepto;
```

Business Problems · Q1–Q4

-- Business Problems

-- Q1. Find the top 10 best-value products based on the discount percentage.

```
select DISTINCT name, mrp, discountPercent
from zepto
order by discountPercent desc
LIMIT 10;
```

-- Q2. What are the Products with High MRP but Out of Stock

```
select DISTINCT name, mrp
from zepto
where outOfStock = TRUE AND mrp > 300
order by mrp desc;
```

-- Q3. Calculate Estimated Revenue for each category

```
select category, SUM(discountedSellingPrice *
availableQuantity) as
total_revenue
from zepto
group by category
order by total_revenue
```

-- Q4. Find all products where MRP is greater than ₹500 and discount is
-- less than 10%.

```
select DISTINCT name, mrp, discountPercent
from zepto
where mrp > 500 AND discountPercent < 10
order by mrp desc, discountPercent desc
```

Bussiness Problems · Q5–Q8

-- Q5. Identify the top 5 categories offering the highest average discount

-- percentage.

```
select category, ROUND(AVG(discountPercent),2) as avg_discount
from zepto
group by category
order by AVG(discountPercent) desc
LIMIT 5;
```

-- Q6. Find the price per gram for products above 100g and sort by best

-- value.

```
select DISTINCT name, weightInGms, discountedSellingPrice,
ROUND((discountedSellingPrice/weightInGms),2) as price_per_grm
from zepto
where weightInGms >=100
order by price_per_grm;
```

--Q7.Group the products into categories like Low, Medium, Bulk.

```
select DISTINCT name, weightInGms,
CASE
    WHEN weightInGms <1000 THEN 'Low'
    WHEN weightInGms <5000 THEN 'Medium'
    ELSE 'Bulk'
END as weight_category
from zepto
```

--Q8.What is the Total Inventory Weight Per Category

```
select category,
SUM(weightInGms * availableQuantity) as total_weight
from zepto
group by category
order by total_weight;
```